



Enhancement of the sensitivity of magneto-optical fiber sensor by magnifying the birefringence of magnetic fluid film with Loyt-Sagnac interferometer

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ABSTRACT

Magnetic field sensor by combining magnetic fluid and optical fiber Loyt-Sagnac interferometer is proposed. The sensor takes advantage of the birefringence effect of magnetic fluid. The relative small birefringence of the magnetic fluid is 'magnified' by the properly designed optical fiber Loyt-Sagnac interferometric structure. As compared to the reported MF-based sensors, the achieved sensitivity of the proposed sensor is 592.8 pm/Oe, which is enhanced by 1–3 orders of magnitude.

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1. Introduction

Magnetic fluid (MF) is one of the most attractive magneto-optical materials, which finds a variety of applications in magneto-optical fiber devices such as magnetic field sensors, magneto-optical modulators, filters, optical switches and so on [1–8]. In these optical devices, MF acting as the magnetically sensitive material modulates the parameters of the light flexibly by interacting with the light directly or indirectly through its various magneto-optical properties such as refractive index tunability, field dependent transmission property, Faraday effect, birefringence effect, dichroism effect, thermal lens effect and so on [9–13]. Especially, MF has the advantages of both the solid magnetic material like strong magnetism and the liquid fluid material like fluidity, which can be employed flexibly in diverse forms in the optical devices. Recently, a lot of research efforts were devoted to improve the sensitivities of MF-based magnetic field sensors by combining MF with various optical fiber measurement configurations. For example, MF films were inserted in the cavity of a Fabry-Perot

interferometer (FPI) or a Sagnac interferometer to realize optical fiber magnetic field sensors [14,15]; MFs were also employed as an outer cladding material of an etched fiber Bragg grating (FBG) [2] or a collapsed PCF based Mach-Zehnder interferometer (MZI) [3], whose sensitivities were 0.344 pm/Oe and 2.367 pm/Oe, respectively; Besides, MF was filled into the air-hole structures of various photonic crystal fibers (PCFs) [1,3,16], whose sensitivities were improved by taking advantage of photonic bandgap effect of PCFs. Though all kinds of sensing configurations were proposed in the reported works, only one magneto-optical effect of MF, saying the tunable refractive index effect, was commonly utilized in the most MF-based magnetic field sensors. However, the tunable range of the refractive index of MF under external magnetic field was quite small (typical value is 0.003 RIU), which limited the sensitivities of the MF-based magnetic field sensors [10,11]. In addition, the concentration of the particles in MF and environmental temperature should be carefully controlled in order to obtain a suitable initial refractive index value. In another aspect, MF possesses a large birefringence coefficient, but also exhibits huge transmission loss due to the increase of the transmission length, so the applications based on the birefringence effect of MF were rarely reported. The Sagnac interferometer has long been used for measuring the magneto-optical effects of magnetic materials [17,18]. In our previous works, we found that the relative small change of birefringence could be magnified and measured by a properly modified Sagnac

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interferometer [19,20]. In this work, we managed to utilize the birefringence effect of MF in a Loyt-Sagnac interferometer to realize a magnetic field sensor with a high sensitivity. A section of polarization maintaining fiber (PMF) was used in the Sagnac loop to introduce a constant phase difference for bias purpose, whilst the MF film was used in the Sagnac loop to introduce variable phase difference for measurement. Meanwhile, the birefringence effect of MF can also be 'magnified' significantly with this proposed optical fiber Loyt-Sagnac interferometric configuration.

2. Magnetic fluid

MF is a translucent suspended colloid of solid nanoscale magnetic particles with the assistance of appropriate surface surfactant coating, which prevents the nanoparticles from agglomerating in the liquid solvent. The ultra fine ferromagnetic nanoparticles disperse by Brownian motion in the liquid carrier with a good physical and optical uniformity and stability. The MF is prepared with the coprecipitation technique [9]. In our experiment, oleic acid capped iron oxide nanoparticles have been used, with nominal particle size of 10 nm. The liquid carrier, organic solvents or water, is the most dominant constituent which governs the overall physical properties of the MF such as viscosity, refractive index, thermal stabilities and so on. In practical experiment, water-based MF is the most common used. The surfactants are very efficient and the MF is stable, which means the magnetic nanoparticles do not agglomerate even in extremely strong magnetic field.

In the absence of an externally applied magnetic field, MF is in an optical homogeneous state, and thus is treated as an optical isotropic material. When the MF is exposed to an externally applied magnetic field, the contained ferromagnetic nanoparticles tend to agglomerate and form chains as well as columns, which align along the external magnetic field direction [21]. Subsequently, the MF comes into a new stable state of balance between the thermal energy and the magnetic energy of the nanoparticles. The agglomeration in the MF causes not only the phase separation between the solid nanoparticles and the liquid solvent, but also the isotropy to the MF as a whole. In this case, the anisotropic MF can be treated as traditional birefringent component with optical axes like an optical uniaxial crystal or a section of PMF. The slow axis of the anisotropic MF is parallel to the magnetic field direction, while the fast axis is perpendicular to the magnetic field direction. The birefringence value of MF depends on the state of balance of the anisotropic MF, and further depends on the externally applied magnetic field strength. The variation trend of the birefringence effect of MF with external magnetic field strength follows the Langevin function [10],

$$B_{MF}(H, T) = (B_s - B_0) \left[\cos(\alpha \frac{H - H_c}{T}) - \frac{T}{\alpha(H - H_c)} \right] + B_0 \quad (1)$$

where H_c is the threshold field strength or critical field strength. B_0 denotes the initial birefringence of the MF when the field strength is lower than the critical field strength; B_s is the saturated birefringence of the MF. α is a coefficient. The Langevin function describes the magnetic column formation in MF which is governed by the thermal energy and magnetic energy [13].

Meanwhile, the structural anisotropy of MF also exhibits dichroism effect, which means light with different polarizations subject to different absorption coefficients. When the externally applied magnetic field is removed, the anisotropic MF returns to the original homogenous state and do not retain magnetization, and thus no birefringence remains. This phenomenon is termed as super-paramagnetism, which is very important for the repeatability of the MF-based magnetic field sensors.

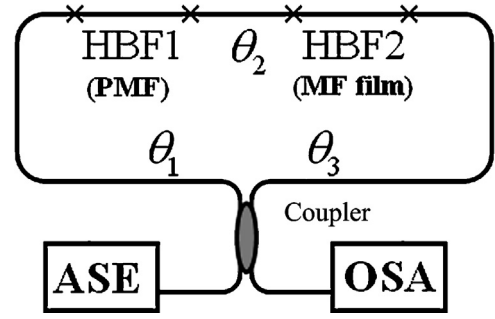


Fig. 1. Schematic diagram of a typical Loyt-Sagnac interferometer.

3. Sensor configuration and principle

A highly birefringent optical fiber Sagnac loop containing two sections of high birefringence fiber (HBF) is referred to as Loyt-Sagnac interferometer or two order Sagnac interferometer, whose typical configuration is shown in Fig. 1. Light launched from the input port of the coupler is split into two beams that counter-propagate along the Sagnac loop. The two beams accumulate phase differences when they pass through the two sections of HBF accordingly and interfere when they beat together at the coupler again. Assuming a 3 dB coupler is used, the relative transmittance of the Loyt-Sagnac interferometer can be obtained with the Jones Matrix method [22],

$$T = \left[\cos \left(\frac{\varphi_1 + \varphi_2}{2} \right) \sin(\theta_1 + \theta_3) \cos \theta_2 + \cos \left(\frac{\varphi_1 - \varphi_2}{2} \right) \cos(\theta_1 + \theta_3) \sin \theta_2 \right]^2 \quad (2)$$

where φ_1 and φ_2 are the phase differences between the two orthogonal modes caused by HBF1 and HBF2 respectively. θ_1 and θ_3 are the angles between the polarization directions of counterpropagating beams and the corresponding fast axes of the two section of HBFs. θ_2 is the angle between the fast axes of the two sections of HBFs. Usually, $\theta_1 + \theta_3$ is controlled by a polarization controller in order to obtain a high fringe contrast, but the polarization controller is not essential.

In the proposed magnetic field sensor based on Loyt-Sagnac interferometric configuration, a section of PMF is in the position of HBF1 while a MF film is in the position of HBF2 with the assistance of a pair of fiber collimators. Besides the birefringence effect of MF film, the dichroism effect is also considered. The fast axes are aligned between the PMF and the MF film ($\theta_2 = 0$) in order to obtain the best sensitivity. The relative transmittance is rededuced as

$$T = 2 \exp[-(\alpha_o + \alpha_e)d] \cdot \sin^2(\theta_1 + \theta_3) \cdot [\cos h(\alpha_o - \alpha_e)d + \cos(\varphi_{PMF} + \varphi_{MF})] \quad (3)$$

where $\varphi_{PMF} = 2\pi B_p L / \lambda$ is the phase difference introduced by the birefringence effect (B_p) of PMF over the fiber length (L); $\varphi_{MF} = 2\pi B_m L / \lambda$ is the phase difference accumulated by the birefringence effect (B_m) of MF film over the thickness (d). α_e and α_o related to the dichroism effect of MF are the absorption coefficients for the two orthogonal modes. The transmission spectrum is an approximately cosinusoidal waveform (Fig. 2) whose fringe contrast depends on the dichroism effect item $\exp[-(\alpha_o + \alpha_e)d]$ and whose period depends on the periodic function item $\cos(\varphi_{PMF} + \varphi_{MF})$. The dip wavelengths on the transmission spectrum can be obtained when the condition $\varphi = \varphi_{PMF} + \varphi_{MF} = (2m + 1)\pi$ (m is an integer) is fulfilled. Hence, the period of the transmission spectrum can be

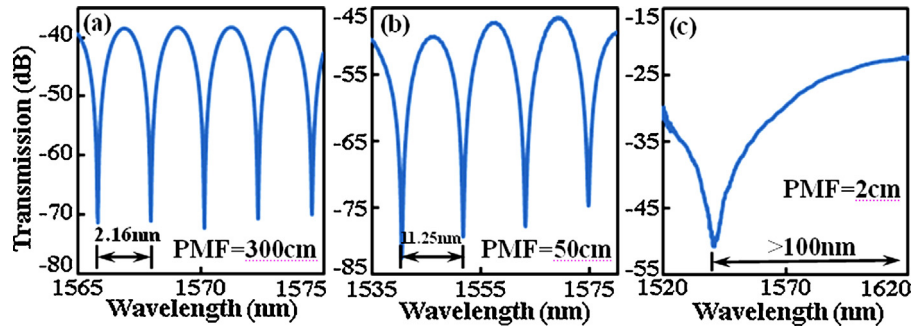


Fig. 2. Transmission spectra of the proposed sensors with different lengths of PMF (a) 300 cm (b) 50 cm (c) 2 cm.

obtained by calculating the wavelength spacing between the adjacent transmission dips,

$$S = \lambda^2 / (B_p L + B_m d) \quad (4)$$

The transmission dips on the spectrum shift with the external magnetic field, therefore, the magnetic field strength can be measured by monitoring the dip wavelengths of the transmission spectrum. The sensitivity of the sensor ($\Delta\lambda$), which is defined by the transmission spectral shift or dip wavelength shift, can be estimated by [23]

$$\Delta\lambda = \frac{\Delta\varphi}{2\pi} S = \frac{\Delta B_m d}{B_p L + B_m d} \lambda \quad (5)$$

Subjected to the huge transmission loss of MF film itself, the thickness of MF film is usually limited to thinner than 100 μm . Therefore, we have $B_p L \gg B_m d$ and Eq. (5) is simplified as,

$$\Delta\lambda \approx \frac{\Delta B_m d}{B_p L} \lambda \quad (6)$$

Eq. (6) shows that the sensitivity of the proposed magnetic field sensor is inversely proportional to the length of PMF (L) and directly proportional to the thickness of MF film (d). In order to enhance the sensitivity, we need to reduce the length of PMF (L), but, in practice, the PMF cannot be shortened without any limitation, because the item $B_m d$ is so small that the period of the transmission spectrum (S) generated solely by MF film itself will be too large and exceed the optical spectral range of the light source according to Eq. (4). Meanwhile, the thickness of MF film cannot be increased without any limitation due to the big optical absorption loss of MF film. If the MF film is too thick, the absorption loss of MF film will be quite high and the OSA cannot detect the output sinusoidal spectrum. The typical thickness of MF film is less than 100 μm .

Briefly, the purpose for introducing PMF into the Sagnac loop is to assist the proposed sensor in producing cosinusoidal transmission spectrum with a suitable period and making the transmission spectrum 'visible' on the measurement devices in the optical communication wavelength range. The sensitivity of the sensor can be enhanced by shortening the length of PMF or increasing the thickness of MF film.

4. Results and discussion

The PMF used in the experiment, which is a kind of HBF, was a traditional pandar fiber (PM-1550-HP), whose birefringence was 3.3×10^{-4} . The water-based MF sample (EMG605 Ferrotech) used in the experiment whose birefringence was estimated to be $\sim 10^{-3}$ [9]. The containment of the MF samples were Fe_3O_4 nanoparticles with nominal diameter of 10 nm and the volume concentration of the nanoparticles was 3.9%. The saturation magnetization, initial magnetic susceptibility and magnetic susceptibility are 220 Gs, 0.24 and 2.96 Gs/Oe, respectively. The MF films were prepared simply

by sealing the MF drops between two 22 mm $\times$ 22 mm glass slides with glue. The thicknesses of the MF film samples were in a range of several tens of microns. In the configuration, the applied magnetic field generated by an electromagnet (EM4-HVA, Lakeshore) was set parallelly to the plane of the MF film while the light path was perpendicular to the plane of MF film. The magnetic field strength was monitored by a Gaussmeter with a transverse probe (Model 425, Lakeshore). An amplified spontaneous emission light source (ASE, 1520–1620 nm) and an optical spectrum analyzer (OSA) were employed to record the transmission spectra of the proposed magnetic field sensors. The experiment was carried out in the room temperature of 24.3 $^\circ\text{C}$.

First, in order to verify the prediction by Eq. (6), three optical fiber Loyt-Sagnac interferometers were prepared by employing the sections of PMF with different lengths. The lengths of the PMFs employed were 300 cm, 50 cm and 2 cm in the three optical fiber Loyt-Sagnac interferometric configurations which are assigned names as Loyt-Sagnac interferometer A, B and C respectively, for convenient description. The MF film sample with a thickness of 40 μm was employed in the three Loyt-Sagnac interferometers and then the magnetic field sensing test was carried out to compare the sensitivities of the three Loyt-Sagnac interferometers. The transmission spectra of the three Loyt-Sagnac interferometers under zero magnetic field are shown in Fig. 2. They are all similar cosinusoidal waveforms with high fringe contrasts (>30 dB). The periods of the transmission spectra are inversely proportional to the PMF-length, which are 2.16 nm, 11.25 nm and more than 100 nm for the Loyt-Sagnac interferometer A, B and C respectively. The largest spectral range of the light source used in the experiment is from 1520 nm to 1620 nm, so there should be at least one dip within this spectral range in order to carry out the measurement. If the length of PMF is too short and the spectral period is too large, it is possible that the period exceeds the spectral range of the light source and we cannot observe any transmission dip in the spectrum. Therefore the shortest length of PMF we chosen was 2 cm in this experiment and its corresponding period of the transmission spectrum was larger than 100 nm.

The variation trends of the transmission spectra of the three Loyt-Sagnac interferometers under external magnetic field are similar. The third transmission dip of the Loyt-Sagnac interferometer B is chosen as an example to describe the variation trend of the transmission spectra (Fig. 3). The transmission spectrum shifts to the longer wavelength side from 1565.45 nm to 1568.58 nm totally 3.13 nm and the fringe contrast declines gradually from 32 dB to 15 dB as the applied magnetic field strength is increased from 0 Oe to 2013 Oe. The decline of the fringe contrast was caused by the dichroism effect of MF, which is predicted in Eq. (3).

The relationships between the dip wavelengths of the three Loyt-Sagnac interferometers and the magnetic field strength are shown in Fig. 4. When the magnetic field strength is smaller than a certain strength (e.g. $H < 200$ Oe), the wavelength shift increases

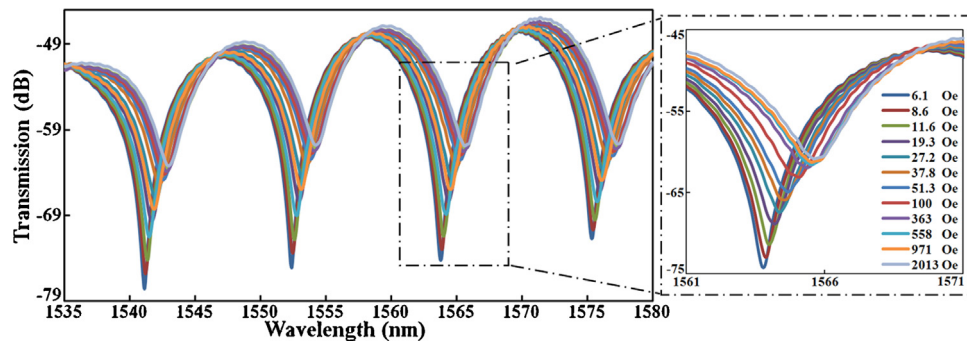


Fig. 3. Transmission spectra of the sensor B under different applied magnetic field strengths.

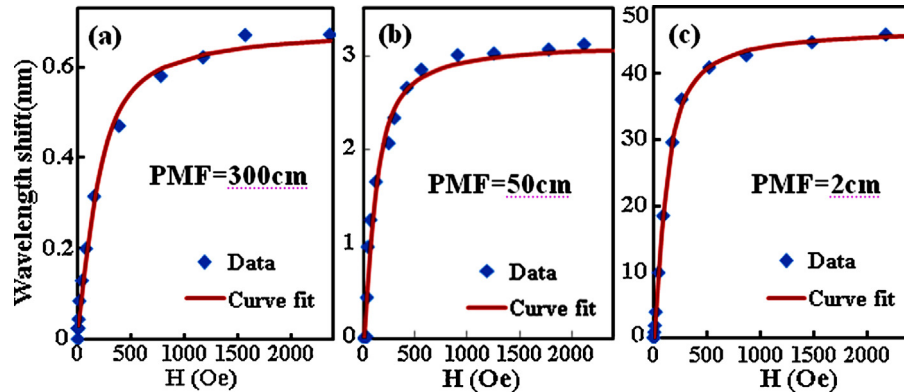


Fig. 4. Dip wavelengths of the transmission spectra vs. the magnetic field strength for the sensors with different PMF-lengths (a) 300 cm (b) 50 cm (c) 2 cm.

Table 1
Comparison the results between Loyt-Sagnac interferometer A, B and C.

	A	B	C
PMF-length (cm)	300	50	2
Spectral Period (nm)	2.16	11.25	>100
Total spectral Shift (nm)	0.67	3.13	46.68
Sensitivity (pm/Oe)	2.0	14.9	218.4

rapidly and linearly with the external magnetic field strength. When the magnetic field strength passes a certain strength (e.g. $H > 500$ Oe), the wavelength shift saturates gradually. Comparing the results of the three Loyt-Sagnac interferometers, the variation trends, which follow a Langevin function, are similar, but the total shifts are different, which are 0.67 nm, 3.13 nm and 46.68 nm for Loyt-Sagnac interferometer A, B and C respectively. Linear fit method is applied to the linear regions of the three curves in Fig. 4. The sensitivities of 2.0 pm/Oe, 14.9 pm/Oe and 218.4 pm/Oe are obtained for Loyt-Sagnac interferometer A, B and C respectively. The results well support the prediction of Eq. (6). Therefore, the birefringence of MF is magnified by using shorter length of PMF in this proposed Loyt-Sagnac interferometric configuration. Meanwhile, MF is superparamagnetic material, there is no hysteresis effect [4].

The key results for Loyt-Sagnac interferometer A, B and C are summarized in Table 1.

In order to further compare and improve the sensitivity performance of the proposed magnetic field sensors, the MF film sample with thickness of $40\ \mu\text{m}$ in Loyt-Sagnac interferometer C was replaced successively by another two MF film samples with corresponding thicknesses of $20\ \mu\text{m}$ and $70\ \mu\text{m}$. The results are shown in Fig. 5. The achieved sensitivities are 126.7 pm/Oe, 218.4 pm/Oe and 592.8 pm/Oe for the MF film samples with thickness of $20\ \mu\text{m}$, $40\ \mu\text{m}$ and $70\ \mu\text{m}$. As for the reported results of the MF-based

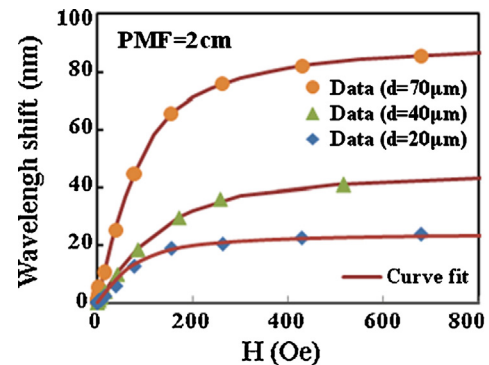


Fig. 5. Dip wavelengths of the transmission spectra vs. the magnetic field strength for the Loyt Sagnac interferometer C with different thicknesses of MF films (a) $20\ \mu\text{m}$ (b) $40\ \mu\text{m}$ (c) $70\ \mu\text{m}$.

magnetic field sensors, the sensitivities of the sensors based on FBG, FPI, MZI and PCF were 0.344 pm/Oe [2], 1.25 pm/Oe [14], 2.37 pm/Oe [5] and 24.2 pm/Oe [1], respectively. Compared with the above results, the achieved sensitivity of this proposed Loyt-Sagnac interferometric configuration (592.8 pm/Oe) is well enhanced.

5. Conclusions

In conclusion, a magnetic field sensor based on Loyt-Sagnac interferometric configuration with an enhanced sensitivity is proposed. The sensor takes advantage of the birefringence effect of MF. In the configuration, the PMF is employed to provide constant phase difference for bias purpose which makes the transmission spectrum visible on the measurement devices whilst the MF film provides variable phase difference for measurement. The

sensitivity performance of the proposed sensor is theoretically analyzed and experimentally verified. The best achieved sensitivity is 592.8 pm/Oe which is 1–3 orders of magnitude better than the reported MF-based sensors. The sensitivity of the proposed also has much room to improve by further reducing the length of the PMF used in the configuration.

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